

The Third Law of Thermodynamics Nobody knows what entropy really is, so in any discussion you will always have to use it.  There are many possible ways of sticking pictures; I can stick them in many different ways.  You can also increase the total entropy of the system by adding more pictures.

Now imagine you are sticking pictures on your wall A , nearby is my wall B . I ask if you could decorate my wall too, so our system $A + B$ has more entropy.  When you start moving pictures to my wall, the total entropy of the system increases.

Inductively, we imagine the universe as a giant closed system, with many particles inside; and they are always exchanging energy with each other.

An ice cube is more "organized" than a cup of water, and vice versa. In the middle of winter, water is frozen. Does this process decrease the entropy of the system?  Surprisingly, the answer is no. The water is cooled down since it loses energy to the surroundings.

But:

Entropy is closely related to energy, but they are not the same, and we do not have any law of conservation of entropy; so we can increase the total entropy of the system.

So the total entropy of our system (a cup of water and the outside environment) increases, because it gains more than it loses.